

Progress Report #2

Project Title: Email Spam Detector: Chrome Extension
with Python Backend

Course: 4495 – Applied Research Project, Section: 002

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Work Logs

Date	Hours Worked	Description of Work Done
Jan 14, 2025	2	Conducted initial research on spam detection techniques, including keyword filtering, heuristic algorithms, Bayesian classifiers, and machine learning. Compiled findings into a research document
Jan 16, 2025	1.5	Researched tools and components required for Chrome extension development. Explored Chrome API documentation and feasibility of integrating with Gmail.
Jan 20, 2025	3	Defined project structure, created a preliminary framework for backend and frontend integration, and discussed Flask API implementation. Documented key design decisions.
Jan 22, 2025	0.5	Created project repository, added initial files and organized folder structure for efficient project management. Encountered difficulty in designing an optimal file structure for seamless integration between frontend and backend.
Jan 25, 2025	1.5	Finalized project proposal, incorporated feedback, and documented technical specifications. Submitted proposal for review.
Jan 27,2025	1.5	Research done on the dataset for project.
Jan 30,2025	2	Started working on project using Jupyter, with the help of Python.
Feb 01,2025	1.5	Faced difficulty in pushing folders on GitHub, tried to figure out the problem.
Feb 05,2025	2.5	Worked together as team to find out errors encountered while data cleaning
Feb 07,2025	2	Worked on pushing files using command prompt and visual studio.
Feb 08,2025	3	Created project report, analysed the work done and challenges faced in project.
Feb 09,2025	2.5	Finalised the project report and submitted for grading.
Feb 12,2025	1.5	Installation of libraries in the independent environment

Feb 17,2025	2	Started work on Front end
Feb 19,2025	3.5	Did model training and worked on front end for user using html and javascript
Feb 23,2025	4	Created reoport and made video for submission.
Feb 24, 2025	2	Researched and analyzed techniques for automatic email detection within a Chrome Extension. Explored DOM parsing and Gmail API integration.
Feb 25, 2025	2.5	Reviewed Chrome Extension security policies and identified potential limitations affecting automatic email extraction. Documented initial findings.
Feb 26, 2025	3	Implemented initial email extraction functionality using DOM parsing. Encountered issues with restricted email access due to security policies.
Feb 27, 2025	3	Tested different ways to bypass security restrictions while complying with Chrome policies. Attempted content script injection to read email content.
Feb 28, 2025	2.5	Attempted to integrate Gmail API for automatic email detection. Faced permission constraints and execution limitations. Documented findings.
Mar 1, 2025	2	Worked on refining the API request structure to handle incoming email data efficiently. Implemented request validation checks.
Mar 2, 2025	3	Improved spam classification model by refining preprocessing techniques for better accuracy. Tested different feature extraction methods.
Mar 3, 2025	2.5	Researched alternative classification models for spam detection. Compared traditional ML (Naïve Bayes, SVM) with deep learning-based approaches (LSTM, CNN).
Mar 4, 2025	2	Developed API endpoints to process extracted email content and return classification results. Ensured efficient request handling.
Mar 5, 2025	3	Conducted integration testing between Chrome Extension and API backend. Fixed minor request-response inconsistencies.
Mar 6, 2025	2.5	Debugged CORS issues affecting communication between the Chrome Extension and backend API. Implemented solutions to improve request-response handling.
Mar 7, 2025	3	Explored UI improvements for the extension, including a clear classification indicator for spam vs. non-spam emails. Added initial styling and layout changes.
Mar 8, 2025	3	Enhanced manual email extraction within the extension to improve user experience. Integrated API call functionality for real-time classification.
Mar 9, 2025	2.5	Conducted speed and performance tests on email classification response time. Optimized API query handling to improve response speed.

Mar 10, 2025	4	Implemented logging mechanisms to track API requests and responses for debugging purposes. Conducted initial tests on backend stability.
Mar 12, 2025	3	Conducted end-to-end testing of the extension and backend integration. Validated classification accuracy and frontend response time.
Mar 13, 2025	2.5	Created detailed documentation on the current system design, listing API endpoints, request-response format, and extension functionality.
Mar 14, 2025	4.5	Documented failed attempts at automatic email detection and explored alternative approaches for future implementation. Planned next steps, including possible use of OAuth for accessing Gmail data securely.
March 15,2025	3	Prepared progress report 2

Description of Work Done

This week, I focused on enhancing the **Chrome Extension and Python backend** for real-time spam detection. My key contributions included:

- **Backend Development:**

- Improved the spam classification model by refining preprocessing techniques and optimizing detection accuracy.
- Developed and tested API endpoints to process email content efficiently.
- Implemented logging mechanisms to track API requests and responses for debugging purposes.

- **Chrome Extension Integration:**

- Worked on automatically detecting emails within the browser without manual selection.
- Faced challenges in implementing automatic email detection due to security restrictions and limitations in accessing email content directly.
- Attempted different approaches, including utilizing DOM parsing and Gmail API, but encountered permission and execution constraints.

- **Testing & Debugging:**

- Conducted tests to validate the extension's ability to extract and classify email content.

- Debugged CORS issues and improved response handling between the extension and the backend.
- Documented failed attempts at automatic email detection and explored potential solutions for future iterations.

Repo Check-In of Implementation:

The following files/folders have been checked into the GitHub repository under the Implementation folder:

Manifest.json – Updated permissions for Gmail access.

content.js – Injected script to extract email data.

background.js – Handles communication with Flask API.

popup.html & popup.js – UI enhancements for manual scanning.

app.py – Flask API for spam detection.

spam_model.pkl – Machine learning model for classification.

Moving forward, we plan to **investigate alternative methods for automating email detection**, optimize backend performance, and improve user experience within the Chrome Extension.